

Companion LII: The Lensing–ISW Cross-Correlation in the Relaxation-Driven Cyclic Cosmology

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Abstract

The cross-correlation between CMB lensing and the integrated Sachs–Wolfe (ISW) effect provides a sensitive probe of the temporal evolution of the gravitational potential. In the standard cosmological model, this correlation arises from the coupling between the time derivative of the potential and the integrated gravitational field along the line of sight. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies both the lensing kernel and the ISW source term, producing a distinctive signature in the lensing–ISW cross-correlation. This Companion develops a continuous description of the cross-correlation in RDCC, deriving how the relaxation scale influences the coupling between lensing and ISW, the coherence of large-scale modes, and the observational signatures in CMB–LSS analyses. The resulting framework provides a direct probe of the temporal structure of the RDCC gravitational field.

1 Introduction

The cross-correlation between CMB lensing and the integrated Sachs–Wolfe (ISW) effect probes the coupling between the spatial and temporal structure of the gravitational potential. CMB lensing is sensitive to the integrated potential $\Phi + \Psi$, while the ISW effect is sensitive to the time derivative $\dot{\Phi} + \dot{\Psi}$. Their cross-correlation therefore encodes the relationship between the potential and its time evolution.

In the standard cosmological model, this correlation is determined by the onset of dark energy domination. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies both the lensing potential and the ISW source term in a scale-dependent manner, producing a distinctive signature in the lensing–ISW cross-correlation.

2 The RDCC Lensing Potential

The CMB lensing potential is given by

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} [\Phi + \Psi](\chi, \hat{\mathbf{n}}). \quad (1)$$

In RDCC, the metric potentials evolve as

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right], \quad (2)$$

$$\Psi_{\text{RDCC}}(k, a) = \Psi(k, a) \left[1 - \chi_{\Psi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

The RDCC lensing potential becomes

$$\phi_{\text{RDCC}} = \phi \left[1 - \chi_\phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

3 The RDCC ISW Source Term

The ISW temperature shift is given by

$$\frac{\Delta T_{\text{ISW}}}{T_\gamma} = 2 \int \dot{\Phi} d\chi. \quad (5)$$

In RDCC, the time derivative of the potential becomes

$$\dot{\Phi}_{\text{RDCC}}(k, a) = \dot{\Phi}(k, a) \left[1 - \chi_\Phi \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right] - \Phi(k, a) \chi_\Phi \alpha \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \frac{\dot{k}_{\text{rel}}}{k_{\text{rel}}}. \quad (6)$$

The second term is unique to RDCC and provides a direct probe of the relaxation mechanism.

4 The Lensing–ISW Cross-Correlation

The cross-correlation is defined as

$$C_L^{\phi T} = \langle \phi_{LM} T_{LM}^* \rangle. \quad (7)$$

In RDCC, this becomes

$$C_L^{\phi T, \text{RDCC}} = C_L^{\phi T} \left[1 - \chi_{\phi T} \left(\frac{L}{L_{\text{rel}}} \right)^\alpha \right] + \Delta C_L^{\phi T, \text{rel}}, \quad (8)$$

where the additional term is

$$\Delta C_L^{\phi T, \text{rel}} \propto \int d\chi \Phi(k, a) \chi_\Phi \alpha \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \frac{\dot{k}_{\text{rel}}}{k_{\text{rel}}}. \quad (9)$$

This term is the hallmark of RDCC.

5 Large-Scale Coherence and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The lensing–ISW cross-correlation therefore exhibits:

- enhanced power at large angular scales,
- suppressed power at small scales,
- a characteristic turnover near L_{rel} ,
- a non-standard phase relation between lensing and ISW.

These signatures provide a direct observational probe of the temporal structure of the RDCC gravitational field.

6 Conclusion

The lensing–ISW cross-correlation in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies both the lensing potential and the ISW source term in a scale-dependent manner, producing distinctive signatures in the cross-correlation spectrum. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB–LSS coupling, potential evolution, and the interplay between matter and gravity in RDCC.

References

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